

# Abhishek Subedi

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## Summary

Graduate Mechanical Engineering student (MS, Dec. 2026) with a focus on thermal-fluid systems and EV/clean energy applications. Strong experimental background in supersonic flow research using PIV and Schlieren imaging. Hands-on experience in battery thermal management, hydrogen fuel cell systems, and heat transfer. Teaching experience across fluid mechanics, thermodynamics, and heat transfer. Published researcher open to thermal engineering, EV powertrain, and R&D roles.

## Education

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| Expected Dec. 2026  | <b>M.S. in Mechanical Engineering</b> , <i>University of Toledo</i> , Toledo, OH                         |
| Aug 2018 – May 2023 | <b>B.E. in Mechanical Engineering</b> , <i>Kathmandu University</i> , Dhulikhel, Nepal<br>GPA: 3.5 / 4.0 |

## Experience

**Graduate Research & Teaching Assistant** *Jan 2025 – Present*  
University of Toledo, Toledo, OH

### Research:

- Experimentally investigating drag coefficients of particles in supersonic flow (*particle-laden supersonic flow*) using Schlieren imaging for flow visualization and Particle Image Velocimetry (PIV) for velocity field characterization; managing test setup, flow conditioning, and data acquisition.
- Designing a valve automation system for a supersonic flow simulator: component selection, control logic, safety interlocks, and technical documentation.

### Teaching:

- Fluid Mechanics Laboratory** – Guided students through experimental setups, data acquisition, and analysis; evaluated lab reports.
- Thermodynamics** – Assisted course instruction, held office hours, and supported grading for undergraduate sections.
- Heat Transfer Laboratory** – Supervised experiments, reinforced theoretical concepts, and ensured safe equipment operation.

**Associate Engineer** *Apr 2024 – Aug 2024*

Mahavir Shree International Pvt. Ltd., Kathmandu, Nepal

- Designed, relocated, and installed baggage handling and mechanical systems for the Domestic Terminal at Tribhuvan International Airport (TIA); coordinated vendors and site teams to minimize operational downtime.
- Supervised scheduled maintenance of Rosenbauer Panther 6x6 fire vehicles at TIA, ensuring operational availability and full compliance with OEM procedures.

**Engineering Intern** *Jan 2023 – Apr 2023*

Sipradi Trading Pvt. Ltd., Lalitpur, Nepal

- Supported service and maintenance of TATA vehicles; completed formal training on engine, transmission, and EV architecture.
- Assisted workflow organization in a high-volume service environment to improve turnaround times and documentation accuracy.

## Projects

### Battery Thermal Management in EVs (Comparative CFD)

- Built CFD models in Ansys to compare active cooling strategies and coolant concentration effects on pack temperature uniformity and peak cell temperatures; assessed trade-offs relevant to EV pack design.

### Hydrogen Fuel Cell Powertrain (Retrofitting a Maruti 800)

- Specified and modeled H<sub>2</sub> fuel cell powertrain integration; performed system-level modeling and safety analysis. Results published in IOP Conference Series (2023).

### Automatic Pipe Cleaning Crawler (Prototype)

- Designed and fabricated a link-and-spring mechanism enabling adaptive radial contact for navigation within variable-diameter pipelines.

## Skills

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| <b>Experimental</b> | PIV; Schlieren imaging; flow measurement; instrumentation & data acquisition; lab setup & validation |
| <b>Simulation</b>   | Ansys Fluent (CFD); Simulink                                                                         |
| <b>CAD</b>          | SolidWorks                                                                                           |
| <b>Programming</b>  | Python (NumPy, Pandas, SciPy, Matplotlib); MATLAB; Git/GitHub; ML (basics)                           |

## Publication

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Y. Shrestha, **A. Subedi**, M. Sapkota, P. Paudel, N. K. Yadav, and B. S. Thapa, “Assessing the performance of a demonstrative hydrogen fuel cell power train in the chassis of an internal combustion engine vehicle,” *IOP Conference Series: Materials Science and Engineering*, vol. 1279, no. 1, p. 012003, Mar. 2023. DOI: 10.1088/1757-899X/1279/1/012003.

## Leadership

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| Aug 2021 – Oct 2022 | <b>Team Leader, Shell Eco-Marathon (Team Junkiri)</b> , <i>Kathmandu University</i> , Dhulikhel, Nepal<br>Led team in vehicle chassis & surface modeling (SolidWorks); coordinated fabrication and international competition logistics (Indonesia). |
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## Training & Certifications

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| May 2023       | <b>Capacity Development Program on EV Technology</b> , Kathmandu University<br>EV components, power electronics, charging systems, efficiency, design/simulation, grid integration. |
| May 2023       | <b>SolidWorks Associate – Mechanical Design</b> , Dassault Systèmes                                                                                                                 |
| Apr – Jul 2024 | <b>Data Science with Python</b> , Broadway Infosys, Kathmandu, Nepal                                                                                                                |